

Seminar 7: Continuous Random Variables. Normal Distribution

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A discrete random variable assigns positive probability to countably many values.

A **continuous random variable** distributes its probability over intervals rather than individual points. In this lecture we study the most important class of continuous distributions: **absolutely continuous distributions**.

1 Probability Distributions

A **probability distribution** on $\mathbb{R}$ is a probability measure

$$P : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1],$$

where $\mathcal{B}(\mathbb{R})$ denotes the Borel σ -algebra.

2 Cumulative Distribution Function

The cumulative distribution function (CDF) of a random variable ξ is defined by

$$F_{\xi}(x) = P(\xi \leq x), \quad x \in \mathbb{R}.$$

Notice that the CDF completely determines the probability distribution.

In other words, if two random variables have the same cumulative distribution function, then they have the same distribution.

3 Properties of the CDF

1. $F(x)$ is non-decreasing.
2. $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.
3. $F(x)$ is right-continuous.